Does Culture Have a Role on the Level of Health Literacy among Underserved Immigrant Population?

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Introduction

Health Literacy and Health Disparity

• Low health literacy limits an individual's ability to communicate with physicians and care providers, implement self-management skills, understand medical conditions, preventive care, and follow medical instructions

• Low levels of health literacy are associated with low levels of education, low levels of income, limited English proficiency, and older age

• Educational levels and English proficiency are the most noticeable factors within Asian immigrants and refugee populations

• Cultural perceptions such as unfamiliarity with the U.S. health care system, cultural norms and expectations surrounding medical care and communication on health add complexity to health literacy levels for immigrants and refugees

Theoretical Framework

• Anderson's Health Behavior Model guided this study

• The three levels of factors to predict health literacy: predisposing factors (such as demographic characteristics and health beliefs), enabling factors (including income and level of health care coverage), and need factors (such as perceived illnesses).

• Anderson is useful in predicting significant variables associated with health literacy (Lee, Choi, & Park, in press; Wister, Malloy-Weir, Rootman, & Desjardins, 2010).

Study Aim

• To investigate the levels of health literacy
• To examine what predisposing, enabling, and need factors are associated with health literacy



Methods

Research Design
• A cross-sectional survey design
• A quota sampling strategy to ensure equal representation of age and gender

Data Collection
• A convenience sample of 407 Korean American immigrants, ages 21 to 90, was recruited in the cities of Corona and Flushing, New York.

• Two data collection methods: (1) face-to-face interviews with participants age 60 and over (2) self-administered questionnaires by those ages 21 to 59

Measures
• Dependent variable: Chew et al's (2004) 16-item health literacy screening scale

• Independent Variables: predisposing factors (gender, age, marital status, and modesty), enabling variables (educational attainment, English proficiency, having a primary care physician), need variables (self-rated health status, number of chronic diseases).

Data Analysis
• Ordinary least squares (OLS) regression analyses

Findings

Health Literacy Profile:
Descriptive Analysis of the Health Literacy Scale (N=407)

Statement	Never-Occasionally		Sometimes		Often-Always	
	N	%	*N*	%	*N*	%
1. How often are appointment slips written in a way that is easy to read and understand?	107	26.3	158	38.8	142	34.9
2. How often are medical forms written in a way that is easy to read and understand?	138	33.9	161	39.6	108	26.5
3. How often are medication labels written in a way that is easy to read and understand?	161	39.6	135	33.2	111	27.3
4. How often are patient educational materials written in a way that is easy to read and understand?	146	35.9	163	40.1	98	24.1
5. How often are appointment slips difficult to understand?	187	46.0	161	39.6	59	14.5
6. How often are medical forms difficult to understand and fill out?	143	35.1	140	34.4	124	30.5
7. How often are directions on medication bottles difficult to understand?	180	44.2	135	33.2	92	22.6
8. How often do you have difficulty understanding written information your health care provider (like doctor, nurse, nurse practitioner) gives you?	167	41.0	140	34.4	100	24.6
9. How often do you have problems getting to your clinic appointments at the right time because of difficulty understanding written instructions?	224	55.0	125	30.7	58	14.3
10. How often do you have problems completing medical forms because of difficulty understanding the instructions?	151	37.1	157	28.6	99	24.3
11. How often do you have problems learning about your medical condition because of difficulty understanding written information?	164	40.4	141	34.7	101	24.9
12. How often are you unsure about how to take your medication(s) correctly because of problems understanding written information on the bottle label?	215	53.0	114	28.1	77	19.0

Multiple Regression Model on Health Literacy of Korean American Immigrants

	Model 1		Model 2		Model 3	
	beta	b	beta	b	beta	b
Predisposing factors						
Female	.097	1.461+	.132	1.968*	.139	2.076*
Age	–.029	–.013	.027	.012	.057	.025
Marital status	–.076	–1.179	–.096	–1.484	–.097	–1.490
Modesty	–.152	–.917**	–.140	–.835**	–.131	–.784*
Enabling factors						
Education attainment			.136	1.077*	.126	1.004*
English proficiency			.176	2.182**	.179	2.211**
Primary care physician			.176	2.797**	.185	2.943**
Need factors						
Health status					.043	.390
Number of chronic diseases					–.082	–1.392
(Constant)		30.966***		20.061***		18.860***
N		375		366		366
Model F		4.47**		7.37***		6.13***
R-square		.046		.126		.134

Note. $*p < .05$; $**p < .01$; $*** p < .001$





Discussion

• More than 60% of the participants reported at the inadequate (31.7%) or marginal (29.8%) levels of health literacy.

•Those with inadequate health literacy struggled with understanding written health care information and filling out medical forms.

•Immigrants who have higher level of modesty when seeking health care services are more likely to be reluctant to pursue or use the services, and in turn are less likely to be informed about or understand health information.

• Of enabling factors, English proficiency, education, and having a primary physician were significant predictors of health literacy

Implications

• Development of patient-centered and culturally competent approach is significant in order to improve quality of care and health outcomes of underserved immigrant populations with limited health literacy.

• A widely affirmed communication strategy is the "teach back" technique, in which health professionals ask patients to repeat in their own words what they understand from the conversation

• One way to address modesty for Korean immigrants would be having a same gender health care provider.

Limitations

• The Korean culture—especially related to health beliefs and care—was not specifically reflected in the 16-item health literacy screening scale used in this study (Chew, Bradley, & Boyko, 2004).

•Low scores on this assessment could reflect unfamiliarity with Western health care concepts and terms (Baker, 2006).

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